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## 3AL Hand-in 1 2024

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**Due: Monday 26 February 11:00 pm**

Please make sure all of your answers are clearly labelled.

1. Determine whether each statement is true or false. No explanation needed.

- (a) If  $H$  and  $K$  are subgroups of  $G$  and  $H \subseteq K$ , then  $H$  is a subgroup of  $K$ .
- (b) If  $H$  and  $K$  are subgroups of  $G$ , then  $H \cap K$  is a subgroup of  $G$ .
- (c) Every homomorphism from a group to itself is an automorphism.
- (d) All homomorphisms preserve the orders of elements.

[4 marks]

2. Prove or disprove: If  $\varphi : G \rightarrow G'$  is a homomorphism and  $H$  is a subgroup of  $G$ , then  $\varphi(H)$  is a subgroup of  $G'$ .

[4 marks]

3. Prove or disprove: If  $\varphi : G \rightarrow G'$  is a homomorphism and  $H \subseteq G$  is such that  $\varphi(H)$  is a subgroup of  $G'$ , then  $H$  is a subgroup of  $G$ .

[4 marks]

Maximum mark = 10

Available marks = 12